AA–TV wiring v9: single IP67 main box (battery inside), 4x M16 cable glands, one solder–board, 1x Deutsch DTP 4–pin (pump). Everything outside the box is flooded ballast space.

BUOY (surface, dry)

BT5
USB–C power bank 5 V / 3 A

J1

Raspberry Pi 4 (buoy)

Pi 4 talks to the Pico over USB serial (/dev/ttyACM0) and powers it from USB 5 V. Reset the Pico by power–cycling the Pi USB ports (uhubctl).

TETHER

W1
USB–A
micro–B
+ USB data

USB–A to micro–B, ~3 m

Vinyl air tube from the buoy is a sheath for the USB cable: it ends on G1. USB plugs are dry (buoy, Pico). A vented box would need a barbed bulkhead instead of a gland.

G1
G1: M16 gland, USB cable

MAIN BOX (IP67, 191 x 111 x 58 mm inside; the only drybox)

BOARD: Pico + servo / signal solder–board (old Boards A + B merged)

Raspberry Pi Pico
A1

220 ohm in series on every PWM line that leaves the board.

TA2
Leak sensor (box low point)

J3
Balance lead in (JST–XH 4)

Board:
– Old Boards A + B merged into one solderable breadboard; no H1 harness.
– 7.4 V + GND rails: 14 AWG soldered along strips; servo + / – jumpers 14 AWG.
– Signal jumpers 20 AWG. No 12 V on the breadboard.
– R13 / R14: 10 k pull–downs so the ESCs see a clean low while the Pico boots.
– Free Pico pins: GP4, GP5, GP17, GP18, GP27 (2nd leak sensor), GP28.

Battery monitor (balance lead –> J3):
– Dividers 47k/100k, 200k/100k, 330k/100k: ~2.9 V at a full pack (4.2 V/cell). ~29 uA per tap.
– Top resistors sit right at J3: a board short only ever sees >= 47 k. Unplug J3 for storage.
– ADS1115 on I2C0 (GP20 SDA, GP21 SCL), addr 0x48. R15/R16: omit if the breakout has pull–ups.

U4
ADS1115 16–bit ADC (breakout)

TB7
7.4 V in (2x14 AWG per pole)

TB2
Rudder servo leads (via G4)

TB3
Elevator servo leads (via G4)

TB4
Thruster ESC signal (U2)

TB5
Pump ESC signal (U3)

Power, off–board (in the box)

BT1–BT3: one 3S LiPo pack, inside the box (cells shown so the balance taps appear). Balance plug (JST–XH 4) –> J3 on the board.

EC5 (pack lead)
EC5 pigtail

Battery + –> EC5 –> F1 –> SW1 –> VBAT bus. Battery – goes straight to the GND bus. High–side switch is required now: J3 ties battery – to board GND, which would bypass a low–side switch. SW1 sits inside the sealed box (lid off to reach it); a reed switch + >= 30 A relay can make it external later.

30 A ATC (standard holder)
Kill switch, HIGH side, >=30 A DC

BB1
+12 V row (OONO 2x6, 30 A)

Bus bar (OONO 2x6, one row per pole, 30 A total):
In: SW1 output / battery –
out: buck U1, thruster ESC U2, pump ESC U3 (2 spare per row)

BB2
GND row (OONO 2x6, 30 A)

U1
12 V –> 7.4 V buck (adjustable, >=15 A)

Set U1 to 7.4 V (Hitec DB777WP max). Savox SW–1250MG: 6.0–8.4 V.

U2
TD1.2 ESC (inside the box)

U2 inside: no 12 V leaves the box and D3 is gone. Only the 3 phase wires go out through G2. ESC outside instead: G2 carries 12 V + GND + PWM + signal GND as a 4–core round cable.

U3
Pump ESC (brushed, reversible, > pump stall)

Pump ESC sits dry, BEC wire cut. Motor pair –> G3 –> D2 –> pump box.

G4
G4: M16 gland, 4 servo leads (insert removed, epoxy–potted)

G2
G2: M16 gland, thruster phase cable

G3
G3: M16 gland, pump 2–core

WET ZONE (flooded hull = ballast): servos, thruster motor, pump cable + D2, pump drybox

Gland rules (M16 x 1.5, check the clamp range on the box listing, often ~4–8 mm):
– A gland seals ONE round cable. Separate wires leak: use round jacketed cable (14/2 for the pump).
– G4: 4 flat servo leads can't seal in a gland. Pull the rubber insert, pass the leads, fill the gland body with epoxy.
– IP67 = 1 m for 30 min. Vacuum–test the closed box before every dive.

Removed vs v8.3: rear drybox, H1 harness (12 x 14 AWG), D1 / D3 / D4 / D5 / D6, TA1 / TB1, rear leak sensor (TB6), the epoxy holes A2 / R2 / R3. Battery, fuse, kill switch and thruster ESC moved inside the box. Fuse holder no longer needs to be waterproof.

Servos: 2x Hitec DB777WP (4.5 A stall @ 7.4 V), 2x Savox SW–1250MG (2.4 A).
Factory leads run uncut into the box through G4 and land on TB2 / TB3.
BACKUP if a lead can't reach: one MGYOO 20 AWG 3–pin waterproof connector per servo.

M4 Rudder servo
M5 Rudder servo
M6 Elevator servo
M7 Elevator servo

MT1
TD1.2 thruster

Factory thruster cable passes G2. Thread the gland nut + seal over it before re–terminating at the ESC.

D2 pump DTP06–4S
DTP04–4P

D2: 1 pump +, 2 pump –; 3–4 dummy loop.

PUMP DRYBOX (or sealed pump)
M3
Ballast pump (12 V, reversible)

High–side kill switch. Servo rail 7.4 V. Cells read by ADS1115 (I2C0).
Glands: G1 USB, G2 thruster phases, G3 pump pair, G4 servo leads (potted). DTP: D2 only.
One IP67 box holds battery, fuse, kill switch, bus bar, buck, both ESCs, one board.

PIE AA–TV

Sheet: /
File: AATV_v9.kicad_sch

Title: AA–TV electrical v9: single IP67 box, 4x M16 glands

Size: A2	Date: 2026–11–06	**Rev: v9**
KiCad E.D.A. kicad–cli 7.0.11+dfsg–1build4		Id: 1/1